

Troca de Chave	Algoritmo	Handshake		Pacotes Trocados	
	Assinatura	Latência [ms]	Amplificação Realizada	Número	Tamanho
BP512 + ML-KEM-1024	P521 + ML-DSA-87	12.8±0.5	3.97x	6 / 15	3778 / 1
BP512 + ML-KEM-1024	P521 + FALCON-1024	14.0±1.2	1.98x	4 / 8	3640 / 1
BP512 + ML-KEM-1024	P521 + FALCON-PADDED-1024	13.0±0.5	1.99x	4 / 8	3640 / 1
BP512 + ML-KEM-1024	P521 + SPHINCS-SHA2-256-F-SIMPLE	36.6±1.5	25.34x	16 / 91	4194 / 1
BP512 + ML-KEM-1024	P521 + SPHINCS-SHA2-256-S-SIMPLE	207.0±0.8	16.56x	9 / 56	3902 / 1
BP512 + ML-KEM-1024	P521 + SPHINCS-SHAKE-256-F-SIMPLE	58.8±1.1	25.60x	15 / 91	4151 / 1
BP512 + ML-KEM-1024	P521 + SPHINCS-SHAKE-256-S-SIMPLE	402.2±1.4	16.22x	11 / 56	3983 / 1
P384 + ML-KEM-1024	P521 + ML-DSA-87	7.9±0.2	3.90x	7 / 16	3849 / 1
P384 + ML-KEM-1024	P521 + FALCON-1024	8.1±0.3	1.97x	4 / 8	3640 / 1
P384 + ML-KEM-1024	P521 + FALCON-PADDED-1024	8.2±0.2	1.98x	4 / 8	3640 / 1
P384 + ML-KEM-1024	P521 + SPHINCS-SHA2-256-F-SIMPLE	29.9±1.3	25.85x	14 / 91	4110 / 1
P384 + ML-KEM-1024	P521 + SPHINCS-SHA2-256-S-SIMPLE	201.3±0.7	16.38x	10 / 56	3942 / 1
P384 + ML-KEM-1024	P521 + SPHINCS-SHAKE-256-F-SIMPLE	50.8±0.4	26.11x	13 / 91	4069 / 1
P384 + ML-KEM-1024	P521 + SPHINCS-SHAKE-256-S-SIMPLE	393.4±1.1	16.38x	10 / 56	3942 / 1
P521 + ML-KEM-1024	P521 + ML-DSA-87	14.9±0.3	3.98x	6 / 15	3778 / 1
P521 + ML-KEM-1024	P521 + FALCON-1024	14.7±0.1	1.98x	4 / 8	3640 / 1
P521 + ML-KEM-1024	P521 + FALCON-PADDED-1024	15.0±0.2	1.99x	4 / 8	3640 / 1
P521 + ML-KEM-1024	P521 + SPHINCS-SHA2-256-F-SIMPLE	35.2±0.3	25.35x	16 / 91	4192 / 1
P521 + ML-KEM-1024	P521 + SPHINCS-SHA2-256-S-SIMPLE	205.8±0.2	16.22x	11 / 56	3983 / 1
P521 + ML-KEM-1024	P521 + SPHINCS-SHAKE-256-F-SIMPLE	57.8±0.4	24.86x	18 / 91	4276 / 1
P521 + ML-KEM-1024	P521 + SPHINCS-SHAKE-256-S-SIMPLE	399.7±1.5	16.42x	10 / 56	3936 / 1
P521 + BIKE-L5	P521 + ML-DSA-87	19.2±0.2	2.53x	10 / 18	7419 / 1
P521 + BIKE-L5	P521 + FALCON-1024	19.8±0.3	1.49x	8 / 12	7337 / 1
P521 + BIKE-L5	P521 + FALCON-PADDED-1024	19.8±0.2	1.50x	8 / 12	7337 / 1
P521 + BIKE-L5	P521 + SPHINCS-SHA2-256-F-SIMPLE	40.0±0.4	13.92x	22 / 94	7902 / 1
P521 + BIKE-L5	P521 + SPHINCS-SHA2-256-S-SIMPLE	211.9±1.2	9.06x	13 / 59	7542 / 1
P521 + BIKE-L5	P521 + SPHINCS-SHAKE-256-F-SIMPLE	64.9±0.5	14.27x	17 / 94	7710 / 1
P521 + BIKE-L5	P521 + SPHINCS-SHAKE-256-S-SIMPLE	404.7±0.9	9.01x	14 / 59	7583 / 1
P521 + FRODO-1344-AES	P521 + ML-DSA-87	22.9±3.9	1.13x	30 / 35	33767 / 1
P521 + FRODO-1344-AES	P521 + FALCON-1024	23.3±3.6	0.94x	29 / 29	32567 / 1
P521 + FRODO-1344-AES	P521 + FALCON-PADDED-1024	20.6±3.8	0.98x	29 / 30	32511 / 1
P521 + FRODO-1344-AES	P521 + SPHINCS-SHA2-256-F-SIMPLE	39.2±0.9	4.14x	34 / 111	31572 / 1
P521 + FRODO-1344-AES	P521 + SPHINCS-SHA2-256-S-SIMPLE	210.0±2.0	2.47x	34 / 75	35060 / 1
P521 + FRODO-1344-AES	P521 + SPHINCS-SHAKE-256-F-SIMPLE	61.7±1.9	3.77x	38 / 110	34099 / 1
P521 + FRODO-1344-AES	P521 + SPHINCS-SHAKE-256-S-SIMPLE	403.7±2.2	2.42x	35 / 77	36290 / 1
P521 + FRODO-1344-SHAKE	P521 + ML-DSA-87	24.2±0.9	1.13x	31 / 36	34967 / 1
P521 + FRODO-1344-SHAKE	P521 + FALCON-1024	23.9±0.5	0.91x	31 / 30	34967 / 1
P521 + FRODO-1344-SHAKE	P521 + FALCON-PADDED-1024	24.2±1.0	0.94x	31 / 32	34911 / 1
P521 + FRODO-1344-SHAKE	P521 + SPHINCS-SHA2-256-F-SIMPLE	54.2±12.8	3.76x	40 / 109	34151 / 1
P521 + FRODO-1344-SHAKE	P521 + SPHINCS-SHA2-256-S-SIMPLE	214.3±0.2	2.47x	35 / 75	35101 / 1
P521 + FRODO-1344-SHAKE	P521 + SPHINCS-SHAKE-256-F-SIMPLE	72.1±2.5	3.63x	41 / 110	35383 / 1
P521 + FRODO-1344-SHAKE	P521 + SPHINCS-SHAKE-256-S-SIMPLE	412.6±6.2	2.47x	35 / 75	35101 / 1
P521 + HQC-256	P521 + ML-DSA-87	143.2±0.4	2.64x	11 / 26	10937 / 1
P521 + HQC-256	P521 + FALCON-1024	150.0±7.8	2.00x	12 / 26	12167 / 1
P521 + HQC-256	P521 + FALCON-PADDED-1024	143.8±0.9	2.00x	12 / 26	12167 / 1
P521 + HQC-256	P521 + SPHINCS-SHA2-256-F-SIMPLE	162.9±0.7	9.64x	19 / 103	12428 / 1
P521 + HQC-256	P521 + SPHINCS-SHA2-256-S-SIMPLE	375.5±11.4	6.95x	18 / 68	11224 / 1
P521 + HQC-256	P521 + SPHINCS-SHAKE-256-F-SIMPLE	197.7±17.0	10.50x	22 / 103	11394 / 1
P521 + HQC-256	P521 + SPHINCS-SHAKE-256-S-SIMPLE	570.0±5.4	6.95x	18 / 68	11224 / 1